

Assignment 2

Technical Documentation

System Overview

The Fire Emergency Volunteer Support Platform is a Power Apps-based emergency response system designed to assist individuals during fire-related emergencies.

The purpose of the platform is to provide a fast and simple way for users to request help during dangerous fire situations. In emergency scenarios, users may experience panic and limited mobility, therefore the system is designed to minimise user interaction while quickly sharing critical information with volunteers and coordinators.

User Roles

The platform supports three primary stakeholders:

- Trapped Individuals — requesting emergency assistance during a fire emergency.
- Fire Stations/ Volunteers — providing assistance and responding to incidents.
- Coordinators — monitoring emergency requests and coordinating volunteer responses.

Core Features

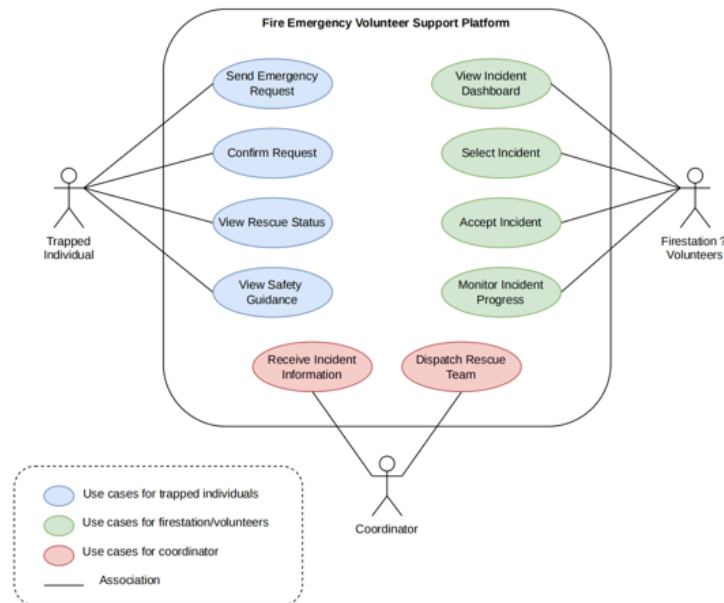
The platform includes several core features:

- Emergency SOS request submission
- Location confirmation and sharing
- Incident monitoring dashboard
- Emergency request tracking

The overall system workflow begins when a trapped individual submits an emergency request through the application. The request information is then stored and shared with nearby fire stations, volunteers, and coordinators, allowing emergency assistance to be organised efficiently.

UML Diagrams

Use Case Diagram



This use case diagram represents the Fire Emergency Volunteer Support Platform and shows how different users interact with the system during an emergency response process.

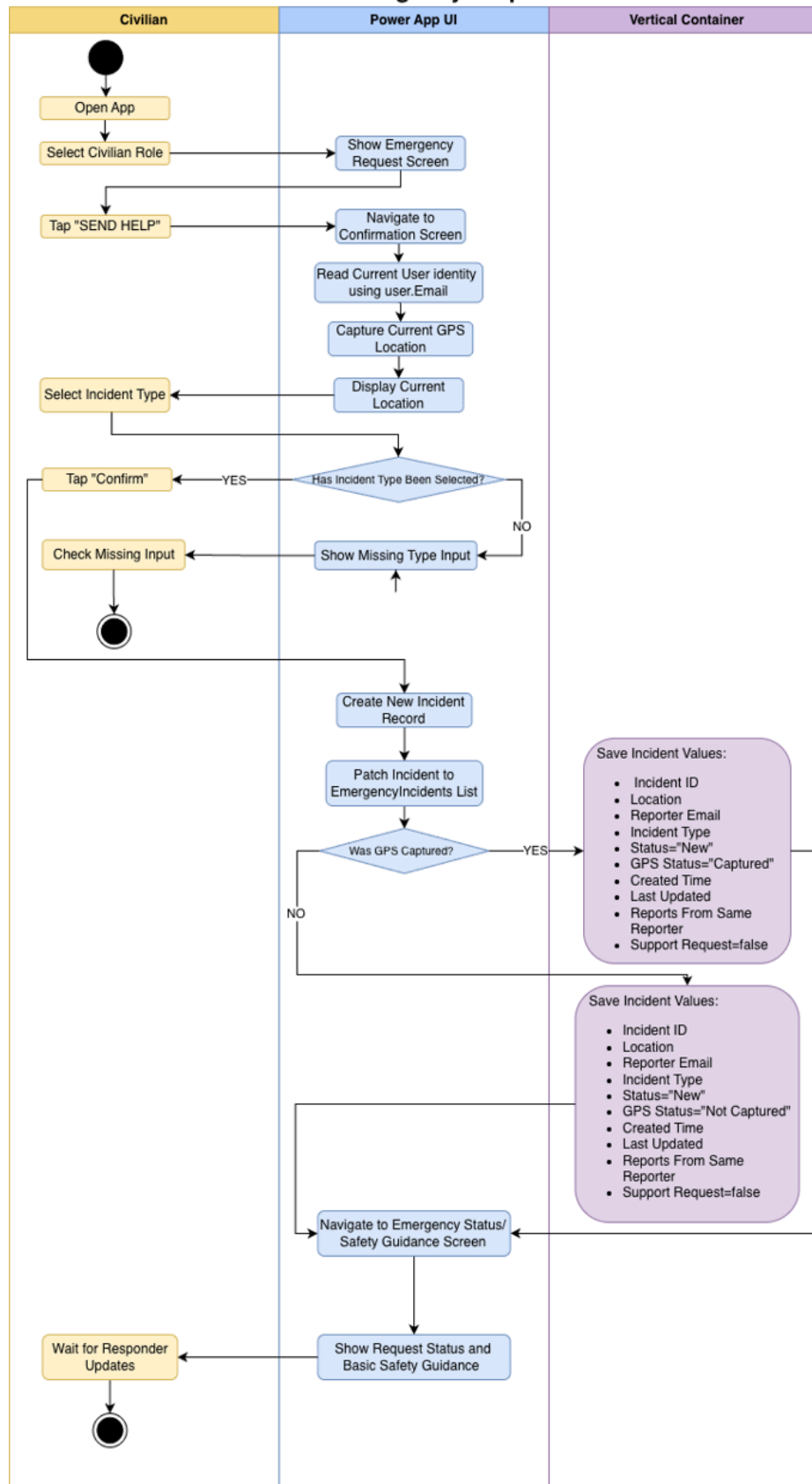
The system contains three main actors:

- Trapped Individual – civilians who request emergency assistance.
- Firestation / Volunteers – responders who manage and respond to incidents.
- Coordinator – responsible for coordinating rescue operations and incident information.

Activity Diagrams

For Civilian:

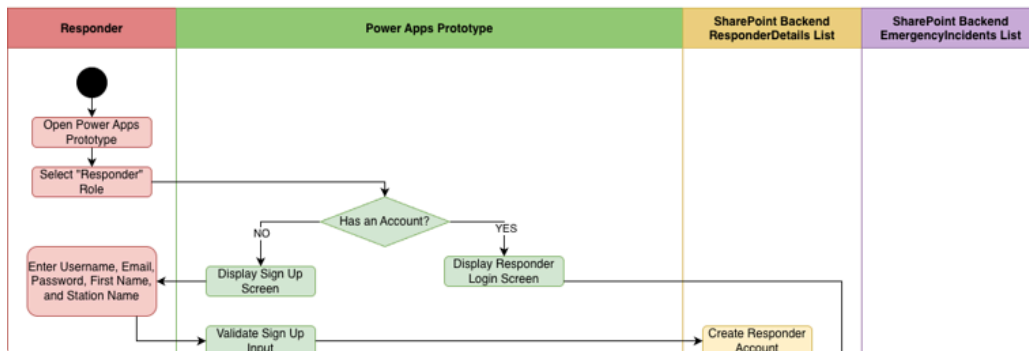
Civilian Emergency Request Flow



This activity diagram illustrates the civilian emergency request workflow within the Power Apps prototype, from submitting an emergency request to receiving responder updates.

For Responder:

Responder Login, Review and Dispatch Flow

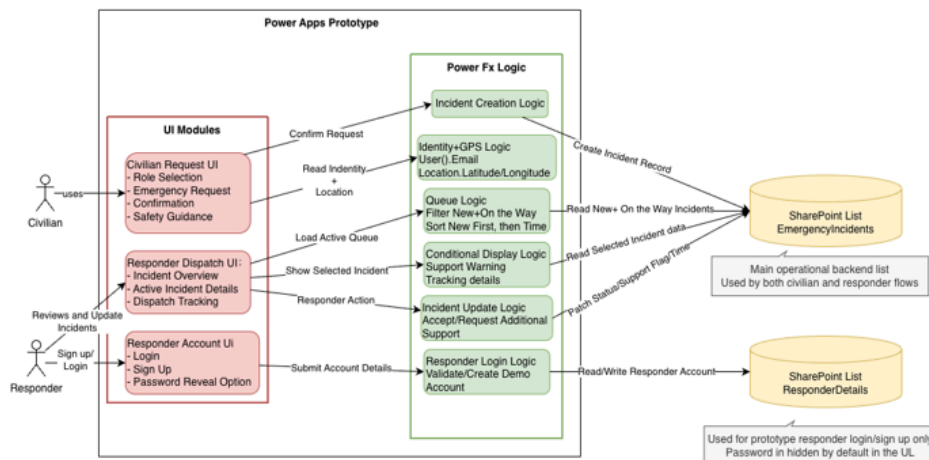


This activity diagram illustrates the responder workflow within the Power Apps prototype, including account registration, login validation, incident review, dispatch actions, and incident tracking.

The workflow begins with responder authentication, where users can either create a new responder account or log in using existing credentials stored in the SharePoint **ResponderDetails** list. After successful login validation, responders are directed to the incident overview screen, where active incidents are loaded from the **EmergencyIncidents** SharePoint list.

System Architecture

System Architecture and SharePoint Backend Overview



This diagram illustrates the overall architecture of the Power Apps prototype and its integration with the SharePoint backend.

The system is divided into three main layers:

- UI Modules
- Power Fx Logic

- SharePoint Backend Lists

The UI modules provide separate interfaces for civilians and responders. Civilian users can submit emergency requests, while responders can review incidents, dispatch support, and update incident statuses.

The Power Fx logic layer handles system behaviour and data processing, including:

- GPS and identity retrieval
- Incident creation
- Queue sorting and filtering
- Incident status updates
- Conditional UI behaviour
- Responder account validation

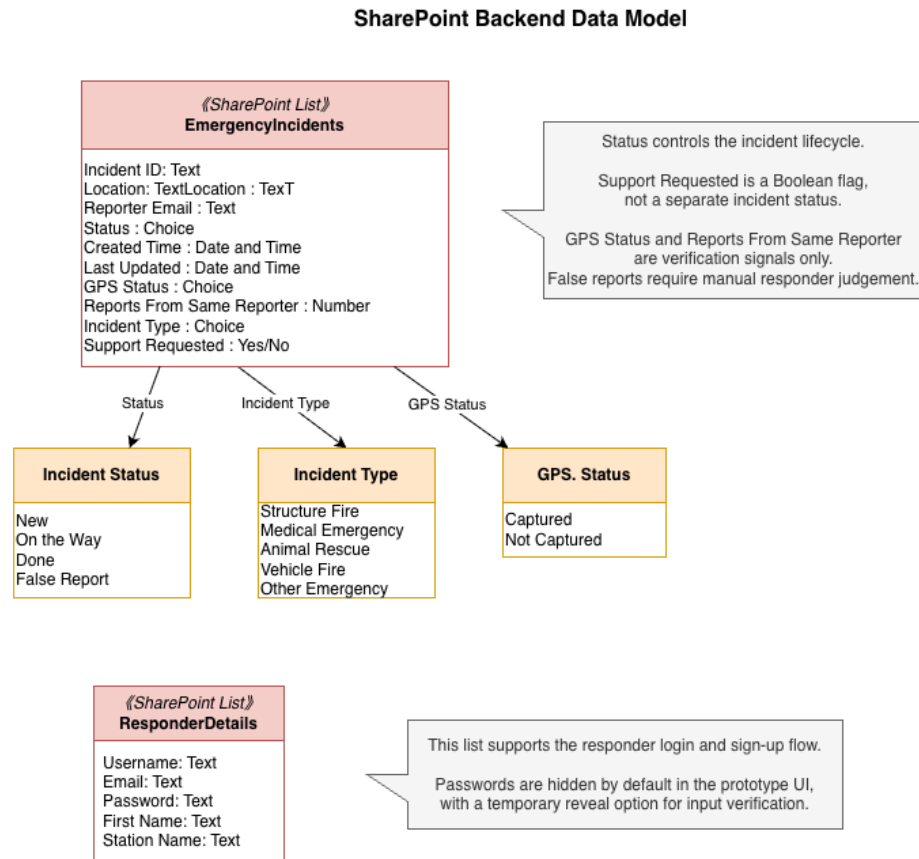
The backend uses two SharePoint lists:

- EmergencyIncidents
- ResponderDetails

The EmergencyIncidents list stores operational incident data used by both civilian and responder workflows. The ResponderDetails list supports the responder login and sign-up prototype flow.

The diagram also demonstrates how data flows between the Power Apps UI, Power Fx logic, and SharePoint backend during incident creation, responder actions, and queue management.

SharePoint Backend Data Model



This diagram illustrates the SharePoint backend data structure used by the Power Apps prototype.

The **EmergencyIncidents** list stores operational incident information, including:

- Incident identifiers
- Reporter information
- Incident types
- Incident status values
- GPS capture status
- Timestamp information
- Support request flags

The backend also defines several controlled value groups, including:

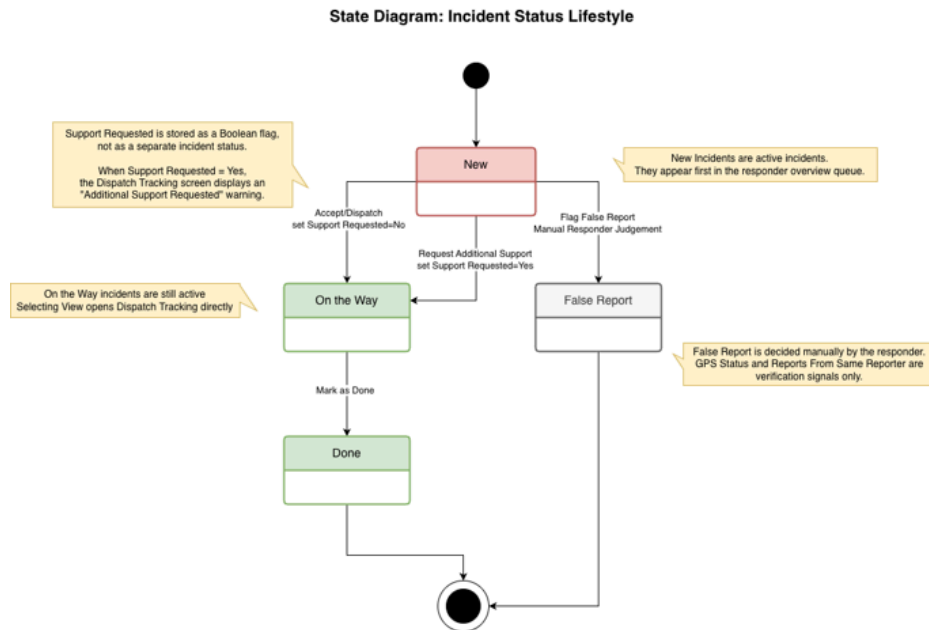
- Incident Status
- Incident Type
- GPS Status

The incident lifecycle is primarily controlled through the **Status** field, while **Support Requested** is implemented as a Boolean flag rather than a separate incident state.

The **GPS Status** field and repeated report counts are used as verification signals only. Final decisions regarding false reports are made manually by responders.

The **ResponderDetails** list supports the responder login and sign-up prototype flow by storing responder account information used during authentication.

State Diagram



This state diagram illustrates the lifecycle of incidents within the emergency response system.

New incidents are first created with the status **New**, where they appear in the responder overview queue as active incidents. Responders may then:

- Accept and dispatch the incident
- Request additional support
- Flag the incident as a false report

Accepted incidents transition to the **On the Way** state, where responders can continue monitoring and updating the incident through the dispatch tracking screen.

When responders mark an incident as completed, the incident transitions to the **Done** state and is removed from the active incident queue.

The system also supports manual false report handling. False reports are determined through responder judgement rather than automatic system detection. GPS status information and repeated reports from the same reporter are treated as verification signals only.

The diagram also demonstrates that **Support Requested** is implemented as a Boolean flag attached to an incident rather than a separate incident status.

Limitations & Future Improvements

Known Limitations

During testing and validation, several limitations were identified in the prototype implementation:

- **Power Apps Test Studio limitations**

Some modern Power Apps controls could not be reliably replayed during automated UI testing.

- **Dynamic gallery validation**

Certain dynamic gallery records required additional manual verification and screenshot evidence instead of fully automated testing.

- **Address conversion restrictions**

Due to connector and geocoding limitations, the system currently displays raw GPS coordinates rather than fully readable addresses.

These limitations were documented together with testing evidence and workarounds during the validation process.

Future Improvements

Several improvements were identified for future versions of the system:

- **Readable address conversion**

Integrate map/geocoding connectors to convert GPS coordinates into user-friendly addresses.

- **Performance optimisation**

Improve SharePoint gallery loading and scalability for larger incident datasets.

- **Accessibility and field usability**

Enhance usability in stressful emergency situations and improve support for mobile field conditions.

- **Stronger responder authentication**

Add stronger authentication and access control mechanisms for responder accounts.

User Guidance

Introduction

This user guide explains how to use the Fire Emergency Volunteer Support Platform.

The platform is designed to support communication and emergency response during fire-related incidents. It allows trapped individuals to quickly request help, while coordinators and fire stations can monitor incidents and coordinate rescue efforts efficiently.

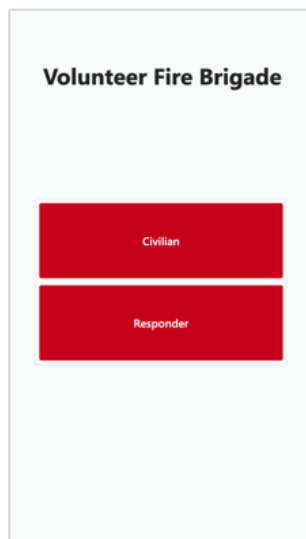
This guide provides step-by-step instructions for the three main user roles within the platform:

- Trapped Individuals
- Coordinators
- Fire Stations

Each section includes simple instructions and screenshots to help users understand the main workflows of the system. For Trapped Individuals

For Civilian

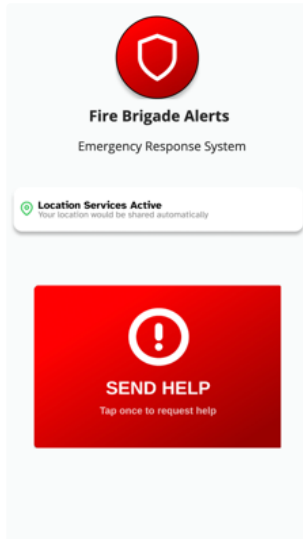
Step 1: Choose Civilian Role



--Tap "Civilian" Here

- Users select the Civilian role to access emergency reporting functions.
- The interface is designed for quick and simple interaction during emergencies

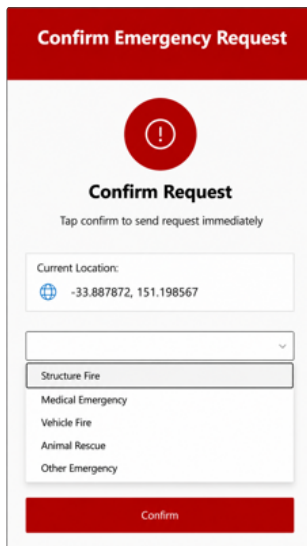
Step 2: Tap SEND HELP



--Tap “SEND HELP” Here

- GPS location is automatically captured.
- Users can quickly submit emergency requests with one tap.

Step 3: Chose Incident Type



--The current GPS location is displayed here

--Tap the Dropdown Arrow

Users can select the different emergency types

Step 4: Confirm Request

Confirm Emergency Request



Confirm Request
Tap confirm to send request immediately

Current Location:
 -33.887872, 151.198567

Structure Fire

Confirm

--Tap "Confirm" Here

- The emergency request will be submitted to the system.
- The request is stored in SharePoint and responders can view the incident in real time.

Step 5: View Rescue Status and Guidance

Fire Brigade Alerts

Current Status:
 New

Safety Guidelines:

- Stay calm and move away from danger
- Do not re-enter the building
- Wait for responder instructions
- Keep your phone nearby

Current Location: -33.88644305, 151.19022653

--View Status Here

--View Guidance Here

For Responder

Step 1: Choose Responder Role

Volunteer Fire Brigade

Civilian

Responder

Step 2: Register or Login

← Back

Volunteer Fire Brigade

Responder Login

Username

Password

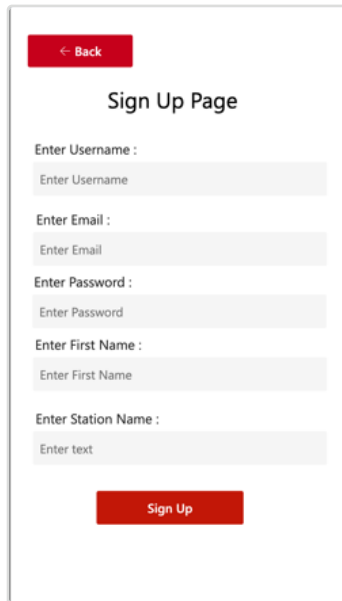


Sign Up

Login

If you do not have an account

--Tap "Sign Up" Here

A mobile app screen titled "Sign Up Page". At the top left is a red button with a white left arrow and the text "Back". Below the title are six input fields, each with a label above it: "Enter Username :", "Enter Email :", "Enter Password :", "Enter First Name :", and "Enter Station Name :". Each input field contains placeholder text matching its label. At the bottom center is a red button with the text "Sign Up".

-- Enter the UserName

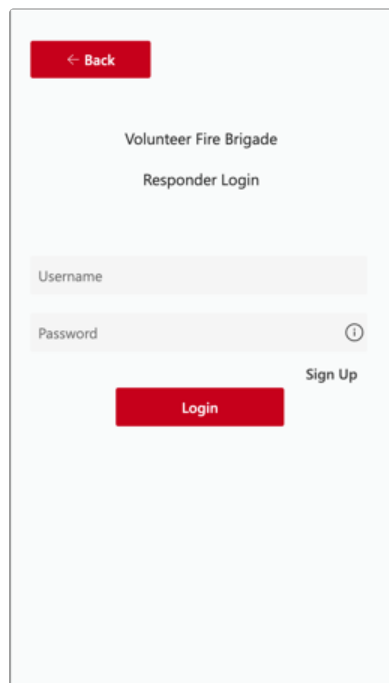
--Enter Email

-- Enter Password

--Enter First Name

--EnterStation Name

--Tap “Sign Up” after entering all information

A mobile app screen titled "Volunteer Fire Brigade" with a subtitle "Responder Login". At the top left is a red button with a white left arrow and the text "Back". Below the title are two input fields: "Username" and "Password". The "Password" field has a small circular icon with an "i" inside to its right. Below the input fields is a red button with the text "Login". To the right of the "Login" button is a link that says "Sign Up".

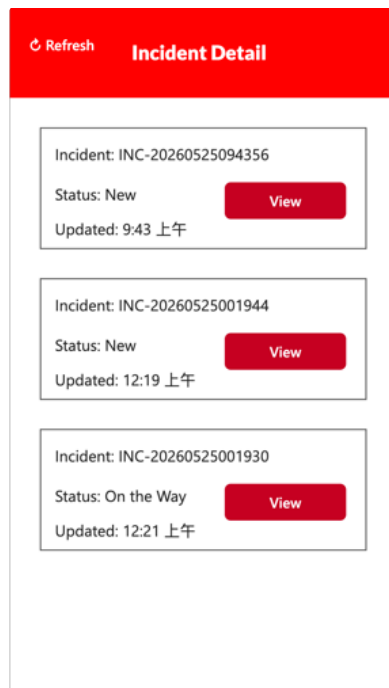
If you already have an account or just registered one, you can log in here

--Input Username Here

--Input Password Here

--Tap “Login” to access the responder dashboard

Step 3: View Dashboard



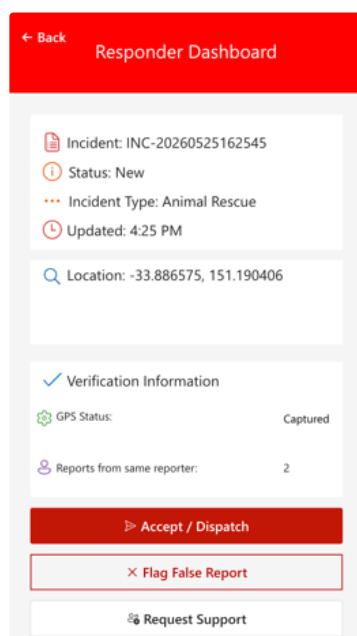
You can view incidents in this page

Incidents are automatically sorted by status and latest updated time.

- New incidents appear first
- “On the Way” incidents remain visible for tracking

--Responders can tap “View” to open incident details

Step 4: View Dashboard



Each incident displays:

- Incident ID
- Status
- Incident Type
- Last Updated Time
- Location

Verification information includes:

- GPS Location Captured or Not
- Reports from Same Reporter

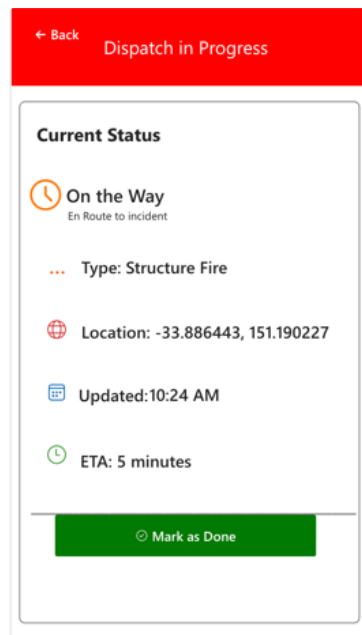
--Tap “Accept/Dispatch” to dispatch responders

--Tap “Flag False Report” if the incident is invalid

--Tap “Request Support” if additional support is needed

If Accept/Dispatch was Chosen:

The incident status is updated to “**On the Way**” and responders can monitor incident progress.



Information displayed:

- Current incident status
- Incident type
- GPS location
- Last updated time
- Estimated arrival time (ETA)

--Tap “Mark as Done” when the incident is resolved

If Flag False Report was Chosen

After flagging a false report, the system returns to the dashboard automatically

- False reports are manually reviewed by responders
- Removed incidents no longer appear in the queue
- Remaining incidents continue to be tracked normally

Refresh

Incident Detail

Incident: INC-20260525001944

Status: New

Updated: 12:19 AM

View

Incident: INC-20260525001930

Status: On the Way

Updated: 12:21 AM

View

If Request Support was Chosen

Back

Dispatch in Progress

Current Status

On the Way

En Route to incident

Warning

Additional Support Requested

Type: Other Emergency

Location: -33.887926, 151.198505

Updated: 10:06 上午

ETA: 8 minutes

Mark as Done

Additional support warning is displayed on the tracking screen.

- “Support Requested” is stored as a Boolean flag
- The incident remains active and visible
- Responders can continue tracking incident progress
- The warning reminds responders that extra resources are needed